Java 核心技术(进阶)

第四章 高级文件处理

第六节 docx简介及解析

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Docx 简介

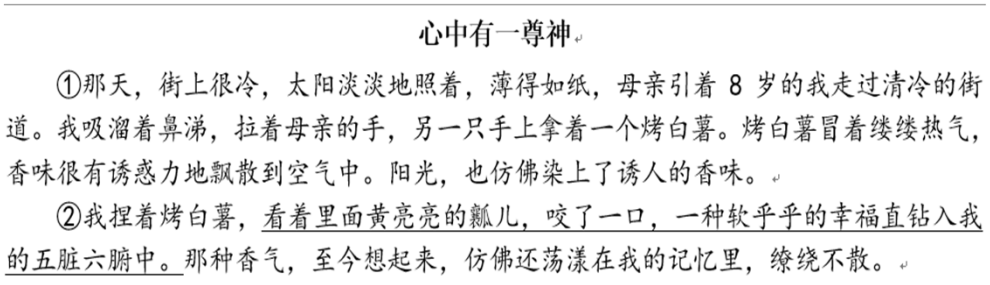
- 以Microsoft Office的doc/docx为主要处理对象。
- Word2003（包括）之前都是doc，文档格式不公开。
- Word2007（包括）之后都是docx，遵循XML路线，文档格式公开。
- docx为主要研究对象
  - 文字样式
  - 表格
  - 图片
  - 公式

Docx 功能和处理

- 常见功能
  - docx解析
  - docx生成（完全生成，模板加部分生成：套打）
- 处理的第三方库
  - Jacob, COM4J (Windows 平台)
  - POI, docx4j, OpenOffice/Libre Office SDK (免费)
  - Aspose (收费)
  - 一些开源的OpenXML的包。

POI

- Apache POI
  - Apache 出品，必属精品， poi.apache.org
  - 可处理docx, xlsx, pptx, visio等office套件
  - 纯Java工具包，无需第三方依赖
  - 主要类
    - XWPFDocument 整个文档对象
    - XWPFParagraph 段落
    - XWPFRun 一个片段(字体样式相同的一段)
    - XWPFPicture 图片
    - XWPFTable 表格



心中有一尊神

①那天，街上很冷，太阳淡淡地照着，薄得如纸，母亲引着 8 岁的我走过清冷的街道。我吸溜着鼻涕，拉着母亲的手，另一只手上拿着一个烤白薯。烤白薯冒着缕缕热气，香味很有诱惑力地飘散到空气中。阳光，也仿佛染上了诱人的香味。

②我捏着烤白薯，看着里面黄亮亮的瓤儿，咬了一口，一种软乎乎的幸福直钻入我的五脏六腑中。那种香气，至今想起来，仿佛还荡漾在我的记忆里，缭绕不散。

Doc/Docx处理总结

- 不同的Office工具，产生出来的docx文件格式不兼容
- 不同的第三方包，能够解析和生成的内容也不同，使用的类也不同
- doc/docx功能非常非常多，第三方包不是万能的，也存在无法解析的情况
- API很多，需要多查询、多练习

代码(1) TextRead.java


```
package docx;

import java.io.FileInputStream;

public class TextRead {

    public static void main(String[] args) throws Exception {
        readDocx();
    }

    public static void readDocx() throws Exception {
        InputStream is;
        is = new FileInputStream("test.docx");
        XWPFDocument xwpf = new XWPFDocument(is);

        List<IBodyElement> ibs= xwpf.getBodyElements();
```

代码(2) TextRead.java

```
for(IBodyElement ib:ibs)
{
    BodyElementType bet = ib.getElementType();
    if(bet== BodyElementType.TABLE)
    {
        //表格
        System.out.println("table" + ib.getPart());
    }
    else
    {
        //段落
        XWPFParagraph para = (XWPFParagraph) ib;
        System.out.println("It is a new paragraph....The indention is "
                + para.getFirstLineIndent() + "," + para.getIndentationFirstLine() );
        //System.out.println(para.getCTP().xmlText());

        List<XWPFRun> res = para.getRuns();
        //System.out.println("run");
        if(res.size()<=0)
        {
            System.out.println("empty line");
        }
```

代码(3) TextRead.java

```
                for(XWPFRun re: res)
                {
                    if(null == re.text()||re.text().length()<=0)
                    {
                        if(re.getEmbeddedPictures().size()>0)
                        {
                            System.out.println("image***" + re.getEmbeddedPictures().size());
                        } else
                        {
                            System.out.println("objects:" + re.getCTR().getObjectList().size());
                            if(re.getCTR().xmlText().indexOf("instrText") > 0) {
                                System.out.println("there is an equation field");
                            }
                            else
                            {
                                //System.out.println(re.getCTR().xmlText());
                            }
                        }
                    }
                    else
                    {
                        System.out.println("==="+ re.getCharacterSpacing() + re.text());
                    }
                }
            }
        }
        is.close();
    }
}
```

代码(4) ImageRead.java

```java
package docx;

 * 本类 完成docx的图片读取工作
import org.apache.poi.openxml4j.exceptions.InvalidFormatException;

public class ImageRead {


    public static void imageRead() throws IOException, InvalidFormatException {
        File docFile = new File("simple.docx");

        XWPFDocument doc = new XWPFDocument(OPCPackage.openOrCreate(docFile));
```

代码(5) ImageRead.java


```
int i = 0;
for (XWPFParagraph p : doc.getParagraphs()) {
    for (XWPFRun run : p.getRuns()) {
        System.out.println("a new run");
        for (XWPFPicture pic : run.getEmbeddedPictures()) {
            System.out.println(pic.getCTPicture().xmlText());
            //image EMU(English Metric Unit)
            System.out.println(pic.getCTPicture().getSpPr().getXfrm().getExt().getCx());
            System.out.println(pic.getCTPicture().getSpPr().getXfrm().getExt().getCy());
            //image 显示大小 以厘米为单位
            System.out.println(pic.getCTPicture().getSpPr().getXfrm().getExt().getCx()/360000.0);
            System.out.println(pic.getCTPicture().getSpPr().getXfrm().getExt().getCy()/360000.0);
            int type = pic.getPictureData().getPictureType();
            byte [] img = pic.getPictureData().getData();
            BufferedImage bufferedImage= ImageRead.byteArrayToImage(img);
            System.out.println(bufferedImage.getWidth());
            System.out.println(bufferedImage.getHeight());
            String extension = "";
```

代码(6) ImageRead.java


```
                switch(type)
                {
                    case Document.PICTURE_TYPE_EMF: extension = ".emf";
                                                    break;
                    case Document.PICTURE_TYPE_WMF: extension = ".wmf";
                        break;
                    case Document.PICTURE_TYPE_PICT: extension = ".pic";
                        break;
                    case Document.PICTURE_TYPE_PNG: extension = ".png";
                        break;
                    case Document.PICTURE_TYPE_DIB: extension = ".dib";
                        break;
                        default: extension = ".jpg";
                }
                //outputFile = new File ( );
                //BufferedImage image = ImageIO.read(new File(img));
                //ImageIO.write(image , "png", outputfile);
                FileOutputStream fos = new FileOutputStream("test" + i + extension);
                fos.write(img);
                fos.close();
                i++;
            }
        }
    }
}
```

代码(7) ImageRead.java


```
    public static BufferedImage  byteArrayToImage(byte[] bytes){
        BufferedImage bufferedImage=null;
        try {
            InputStream inputStream = new ByteArrayInputStream(bytes);
            bufferedImage = ImageIO.read(inputStream);
        } catch (IOException ex) {
            System.out.println(ex.getMessage());
        }
        return bufferedImage;
}
    public static void main(String[] args) throws Exception {
        imageRead();
    }
}
```

代码(8) ImageWrite.java


```
package docx;

 * 本类完成docx的图片保存工作
import org.apache.poi.util.Units;


public class ImageWrite {

    public static void main(String[] args) throws Exception {
        XWPFDocument doc = new XWPFDocument();
        XWPFParagraph p = doc.createParagraph();

        XWPFRun r = p.createRun();

        String[] imgFiles = new String[2];

        imgFiles[0] = "c:/temp/ecnu.jpg";
        imgFiles[1] = "c:/temp/shida.jpg";
```

代码(9) ImageWrite.java


```
for(String imgFile : imgFiles) {
    int format;

    if(imgFile.endsWith(".emf")) format = XWPFDocument.PICTURE_TYPE_EMF;
    else if(imgFile.endsWith(".wmf")) format = XWPFDocument.PICTURE_TYPE_WMF;
    else if(imgFile.endsWith(".pict")) format = XWPFDocument.PICTURE_TYPE_PICT;
    else if(imgFile.endsWith(".jpeg") || imgFile.endsWith(".jpg")) format = XWPFDocument.PICTURE_TYPE_JPEG;
    else if(imgFile.endsWith(".png")) format = XWPFDocument.PICTURE_TYPE_PNG;
    else if(imgFile.endsWith(".dib")) format = XWPFDocument.PICTURE_TYPE_DIB;
    else if(imgFile.endsWith(".gif")) format = XWPFDocument.PICTURE_TYPE_GIF;
    else if(imgFile.endsWith(".tiff")) format = XWPFDocument.PICTURE_TYPE_TIFF;
    else if(imgFile.endsWith(".eps")) format = XWPFDocument.PICTURE_TYPE_EPS;
    else if(imgFile.endsWith(".bmp")) format = XWPFDocument.PICTURE_TYPE_BMP;
    else if(imgFile.endsWith(".wpg")) format = XWPFDocument.PICTURE_TYPE_WPG;
    else {
        System.err.println("Unsupported picture: " + imgFile +
                ". Expected emf|wmf|pict|jpeg|png|dib|gif|tiff|eps|bmp|wpg");
        continue;
    }
```

代码(10) ImageWrite.java


```
            r.setText(imgFile);
            r.addBreak();
            r.addPicture(new FileInputStream(imgFile), format, imgFile, Units.toEMU(200), Units.toEMU(200)); // 200x200 pixels
            r.addBreak(BreakType.PAGE);
        }

        FileOutputStream out = new FileOutputStream("images.docx");
        doc.write(out);
        out.close();
    }


}
```

代码(11) TableRead.java


```java
package docx;

 * 本类完成docx的表格内容读取
import java.io.FileInputStream;


public class TableRead {
    public static void main(String[] args) throws Exception {
        testTable();
    }

    public static void testTable() throws Exception {
        InputStream is = new FileInputStream("simple2.docx");
        XWPFDocument xwpf = new XWPFDocument(is);
        List<XWPFParagraph> paras = xwpf.getParagraphs();
        //List<POIXMLDocumentPart> pdps = xwpf.getRelations();
```

代码(12) TableRead.java


```
List<IBodyElement> ibs= xwpf.getBodyElements();
for(IBodyElement ib:ibs)
{
    BodyElementType bet = ib.getElementType();
    if(bet== BodyElementType.TABLE)
    {
        //表格
        System.out.println("table" + ib.getPart());
        XWPFTable table = (XWPFTable) ib;
        List<XWPFTableRow> rows=table.getRows();
         //读取每一行数据
        for (int i = 0; i < rows.size(); i++) {
            XWPFTableRow  row = rows.get(i);
            //读取每一列数据
            List<XWPFTableCell> cells = row.getTableCells();
            for (int j = 0; j < cells.size(); j++) {
                XWPFTableCell cell=cells.get(j);
                System.out.println(cell.getText());
                List<XWPFParagraph> cps = cell.getParagraphs();
                System.out.println(cps.size());
            }
        }
    }
```

代码(13) TableRead.java


```
else
{
    //段落
    XWPFParagraph para = (XWPFParagraph) ib;
    System.out.println("It is a new paragraph....The indention is "
       + para.getFirstLineIndent() + "," + para.getIndentationFirstLine() + ","
       + para.getIndentationHanging()+"," + para.getIndentationLeft() + ","
       + para.getIndentationRight() + "," + para.getIndentFromLeft() + ","
       + para.getIndentFromRight()+"," + para.getAlignment().getValue());

    //System.out.println(para.getAlignment());
    //System.out.println(para.getRuns().size());

    List<XWPFRun> res = para.getRuns();
    System.out.println("run");
    if(res.size()<=0)
    {
        System.out.println("empty line");
    }
```

代码(14) TableRead.java


```
                for(XWPFRun re: res)
                {
                    if(null == re.text()||re.text().length()<=0)
                    {
                        if(re.getEmbeddedPictures().size()>0)
                        {
                            System.out.println("image***" + re.getEmbeddedPictures().size());

                        }
                        else
                        {
                            System.out.println("objects:" + re.getCTR().getObjectList().size());
                            System.out.println(re.getCTR().xmlText());

                        }
                    }
                    else
                    {
                        System.out.println("===" + re.text());
                    }
                }

            }
        }
        is.close();
    }
}
```

代码(15) TableWrite.java


```
package docx;

 * 本类测试写入表格

import java.io.FileOutputStream;

public class TableWrite {

 public static void main(String[] args) throws Exception {
    try {
        createSimpleTable();
    }
    catch(Exception e) {
        System.out.println("Error trying to create simple table.");
        throw(e);
    }
 }
```

代码(16) TableWrite.java


```
public static void createSimpleTable() throws Exception {
    XWPFDocument doc = new XWPFDocument();

    try {
        XWPFTable table = doc.createTable(3, 3);

        table.getRow(1).getCell(1).setText("表格示例");

        XWPFParagraph p1 = table.getRow(0).getCell(0).getParagraphs().get(0);

        XWPFRun r1 = p1.createRun();
        r1.setBold(true);
        r1.setText("The quick brown fox");
        r1.setItalic(true);
        r1.setFontFamily("Courier");
        r1.setUnderline(UnderlinePatterns.DOT_DOT_DASH);
        r1.setTextPosition(100);
```

代码(17) TableWrite.java


```
            table.getRow(2).getCell(2).setText("only text");

            OutputStream out = new FileOutputStream("simpleTable.docx");
            try {
                doc.write(out);
            } finally {
                out.close();
            }
        } finally {
            doc.close();
        }
    }
}
```

代码(18) TemplateTest.java


```
public class TemplateTest {

    public static void main(String[] args) throws InvalidFormatException, FileNotFoundException, IOException {
        XWPFDocument doc = openDocx("template.docx");//导入模板文件
        Map<String, Object> params = new HashMap<>();//文字类 key-value
        params.put("${name}", "Tom");
        params.put("${sex}", "男");
        Map<String,String> picParams = new HashMap<>();//图片类 key-url
        picParams.put("${pic}", "c:/temp/ecnu.jpg");
        List<IBodyElement> ibes = doc.getBodyElements();
        for (IBodyElement ib : ibes) {
            if (ib.getElementType() == BodyElementType.TABLE) {
                replaceTable(ib, params, picParams, doc);
            }
        }
        writeDocx(doc, new FileOutputStream("template2.docx"));//输出
    }
```

代码(19) TemplateTest.java


```
public static XWPFDocument openDocx(String url) {
    InputStream in = null;
    try {
        in = new FileInputStream(url);
        XWPFDocument doc = new XWPFDocument(in);
        return doc;
    } catch (FileNotFoundException e) {
        e.printStackTrace();
    } catch (IOException e) {
        e.printStackTrace();
    } finally {
        if (in != null) {
            try {
                in.close();
            } catch (IOException e) {
                // TODO Auto-generated catch block
                e.printStackTrace();
            }
        }
    }
    return null;
}
```

代码(20) TemplateTest.java


```
public static void writeDocx(XWPFDocument outDoc, OutputStream out) {
    try {
        outDoc.write(out);
        out.flush();
        if (out != null) {
            try {
                out.close();
            } catch (IOException e) {
                e.printStackTrace();
            }
        }
    } catch (IOException e) {
        e.printStackTrace();
    }
}

private static Matcher matcher(String str) {
    Pattern pattern = Pattern.compile("\\$\\{(.+?)\\}", Pattern.CASE_INSENSITIVE);
    Matcher matcher = pattern.matcher(str);
    return matcher;
}
```

代码(21) TemplateTest.java


```
public static void replacePic(XWPFRun run,  String imgFile,  XWPFDocument doc) throws Exception {
    int format;
    if (imgFile.endsWith(".emf"))
        format = Document.PICTURE_TYPE_EMF;
    else if (imgFile.endsWith(".wmf"))
        format = Document.PICTURE_TYPE_WMF;
    else if (imgFile.endsWith(".pict"))
        format = Document.PICTURE_TYPE_PICT;
    else if (imgFile.endsWith(".jpeg") || imgFile.endsWith(".jpg"))
        format = Document.PICTURE_TYPE_JPEG;
    else if (imgFile.endsWith(".png"))
        format = Document.PICTURE_TYPE_PNG;
    else if (imgFile.endsWith(".dib"))
        format = Document.PICTURE_TYPE_DIB;
    else if (imgFile.endsWith(".gif"))
        format = Document.PICTURE_TYPE_GIF;
    else if (imgFile.endsWith(".tiff"))
        format = Document.PICTURE_TYPE_TIFF;
    else if (imgFile.endsWith(".eps"))
        format = Document.PICTURE_TYPE_EPS;
    else if (imgFile.endsWith(".bmp"))
        format = Document.PICTURE_TYPE_BMP;
    else if (imgFile.endsWith(".wpg"))
        format = Document.PICTURE_TYPE_WPG;
```

代码(22) TemplateTest.java


```
    else {
        System.err.println(
                "Unsupported picture: " + imgFile + ". Expected emf|wmf|pict|jpeg|png|dib|gif|tiff|eps|bmp|wpg");
        return;
    }
    if(imgFile.startsWith("http")||imgFile.startsWith("https")){
        run.addPicture(new URL(imgFile).openConnection().getInputStream(), format, "rpic",Units.toEMU(100),Units.toEMU(100));
    }else{
        run.addPicture(new FileInputStream(imgFile), format, "rpic",Units.toEMU(100),Units.toEMU(100));
    }
}
```

代码(23) TemplateTest.java


```
public static void replaceTable(IBodyElement para ,Map<String, Object> params,
        Map<String, String> picParams, XWPFDocument indoc)
        throws Exception {
    Matcher matcher;
    XWPFTable table;
    List<XWPFTableRow> rows;
    List<XWPFTableCell> cells;
    table = (XWPFTable) para;
    rows = table.getRows();
    for (XWPFTableRow row : rows) {
        cells = row.getTableCells();
        int cellsize = cells.size();
        int cellcount = 0;
        for(cellcount = 0; cellcount<cellsize;cellcount++){
            XWPFTableCell cell = cells.get(cellcount);
            String runtext = "";
            List<XWPFParagraph> ps = cell.getParagraphs();
            for (XWPFParagraph p : ps) {
                for(XWPFRun run : p.getRuns()){
                    runtext = run.text();
                    matcher = matcher(runtext);
```

代码(24) TemplateTest.java


```
                    if (matcher.find()) {
                        if (picParams != null) {
                            for (String pickey : picParams.keySet()) {
                                if (matcher.group().equals(pickey)) {
                                    run.setText("",0);
                                    replacePic(run, picParams.get(pickey), indoc);
                                }
                            }
                        }
                        if (params != null) {
                            for (String pickey : params.keySet()) {
                                if (matcher.group().equals(pickey)) {
                                    run.setText(params.get(pickey)+"",0);
                                }
                            }
                        }
                    }
                }
            }
        }
    }
}
```

谢 谢！